

# Hao Ai

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## EDUCATION

**University of Birmingham**, Birmingham, UK **September 2024 - Now**  
**PhD.** in Computer Science (supervised by Prof. Jianbo Jiao, Prof. Ales Leonardis, and Prof. Eyal Ofek)  
**Research Focus:** 3D vision, vision SLAM, scene understanding, 360 vision

**Tsinghua University**, Beijing, China **September 2018- June 2021**  
**M.S.** in Electronic Engineering (supervised by Prof. Qingmin Liao)  
**Main Course:** Stochastic Processes, Pattern Recognition, Statistical Signal Processing, Digital Image Processing, etc.  
**GPA:** 3.29/4.0  
**Research Focus:** Face recognition, image classification, and representation learning  
**Master Thesis:** Feature Based Loss Functions on Image Classification Task with CNN Approaches (Excellent Graduation Thesis candidate)

**Beijing Jiaotong University**, Beijing, China **September 2014- June 2018**  
**B.S.** in Communication Engineering (work with Prof. Yuchun Guo)  
**Main Course:** Communications Principle, Linear Algebra, Calculus, Digital Signal Processing, etc.  
**Bachelor Thesis:** Research on Relationships between Customer Comments and Sales on E-commerce Platform (*based on machine learning*)

## RESEARCH INTEREST

I am deeply passionate about signal processing, deep learning, and computer vision. The world of images, serving as two-dimensional snapshots of reality, has always fascinated me. I am particularly intrigued by the prospect of extracting three-dimensional insights from these flat pictures.

After my Master's degree, I focused on 360° images captured by novel panoramic camera sensors that provide a wide field of view. From a single 360° image together with an estimated depth map, it is possible to reconstruct a relatively complete 3D scene of the surrounding environment.

In my PhD, I seek to explore the potential of neural 3D representations for dynamic scenes, *e.g.*, 3D Gaussian Splatting and NeRF. I am particularly interested in leveraging 2D motion cues in the observational space to model and perceive the 3D world, and using human-world interaction to enable interactive 3D world generation.

## SKILLS

Proficient in deep learning frameworks such as PyTorch and TensorFlow, I also possess valuable experience in MATLAB and CUDA. Moreover, during my industry journey, I gained experience in JAVA.

## PUBLICATIONS *First Author or Co-first author*

**H. Ai**, W. Chang, J. Jiao, A. Leonardis, E. Ofek. Articulation in Motion: Prior-free Part Mobility Analysis for Articulated Objects By Dynamic-Static Disentanglement. *ICLR 2026*

Z. Cao\*, J. Zhu\*, **H. Ai\***, L. Jiang, Y. Lyu, H. Xiong. ST<sup>2</sup>360D: Spatial-to-Temporal Consistency for Training-free 360 Monocular Depth Estimation. *NIPS 2025*

**H. Ai**, Z. Cao, L. Wang. A survey of representation learning, optimization strategies, and applications for omnidirectional vision. *International Journal of Computer Vision 2025*

**H. Ai**, Z. Cao, H. Lu, C. Chen, J. Ma, P. Zhou, T. Kim, P. Hui, L. Wang. Dream360: Diverse and Immersive Outdoor Virtual Scene Creation via Transformer-Based 360 Image Outpainting. *IEEE VR 2024 & IEEE Transactions on Visualization and Computer Graphics*

**H. Ai**, L. Wang. Elite360D: Towards Efficient 360 Depth Estimation via Semantic- and Distance-Aware Bi-Projection Fusion. *CVPR 2024*

**H. Ai**, Z. Cao, YP. Cao, Y. Shan, L. Wang. HRDFuse: Monocular 360° Depth Estimation by Collaboratively Learning Holistic-With-Regional Depth Distributions. *CVPR 2023*

### *Others*

Z. Cao, J. Zhu, W. Zhang, **H. Ai**, H. Bai, H. Zhao, L. Wang. PanDA: Towards Panoramic Depth Anything with Unlabeled Panoramas and Mobius Spatial Augmentation. *CVPR 2025*

W. Chang, **H. Ai**, L. Wang. CUBE360: Learning Cubic Field Representation for Monocular Panoramic Depth Estimation

Z. Cao, **H. Ai**, YP. Cao, Y. Shan, X. Qie, L. Wang. OmniZoomer: Learning to Move and Zoom in on Sphere at High-Resolution. *ICCV 2023*

## RESEARCH EXPERIENCE

**Research assistant, HKUST(GZ)** with Prof. Addison Lin Wang **May 2022-Aug 2024**  
& **Internship in ARCLab (Tencent)** with Dr. Yan-Pei Cao **May 2022- February 2023**

Worked on *HRDFuse*, *Elite360D*, *Dream360*, *A survey for omnidirectional vision* as the first author, *OmniZoomer* as the second author.

- (*In Tencent ARCLab*) Defined the research problem of fully leveraging the multiple projections of 360° images, proposed **HRDFuse** to combine holistic contextual information and regional structural information, designed and conducted experiments, analyzed the results, and wrote the paper.
- Further investigated bi-projection fusion for 360° depth estimation and introduced **Elite360D**. This approach utilizes ERP images and an icosahedron projection point set as inputs for an efficient fusion model.
- Inspired by auto-regressive transformer-based generative models, we proposed **Dream360**, a generative model specifically designed for 360 images. Dream360 learns a sphere-specific code-book, outpaints freely masked panoramas, and refines the generated outputs. I designed and conducted the experiments, analyzed the results, and wrote the paper.
- Contributed to the idea of OmniZoomer, combining the Mobius transformation and 360° image super-resolution to produce HR and high-quality ODIs with the flexibility to move and zoom in to the object of interest, and wrote the paper.
- Proposed the first systematic and comprehensive review and analysis of the recent progress in deep learning methods for omnidirectional vision with my partners. We analyzed over 200 high-quality papers and provided an insightful discussion of the challenges and open problems.

**Tsinghua VIP Lab** with Prof. Qingmin Liao

**September 2018- June 2021**

**Internship in Ping An Technology**

**March 2020- January 2021**

Worked on *Obstructive Sleep Apnea detection* project and *Robust Face Recognition* project.

- Worked with an undergraduate student of computer science and a doctor of Medicine to accomplish the Obstructive Sleep Apnea detection, which is a sub-project of a National Key Research and Development Program of China.
- Contributed to the research ideation by extending the object detection method for images into the 1-D sound signals, designed and ran the experiments, and wrote the paper.
- (*In Ping An Technology*) Proposed a VAE model based on multivariate Gaussian mixture distributions for face recognition, designed and ran the experiments, and wrote the paper. The performance of proposed model has proven to be significantly improved on multiple benchmarks.

## INDUSTRY EXPERIENCE

**Algorithms Engineer in Beijing Research Institute of Huawei Technology Co., Ltd**

July 2021-March 2022

- Designed and developed algorithms for autonomous driving network (ADN), which introduces the power of artificial intelligence into the management of communication networks for more effectively and efficiently detecting and solving the network error.